

RESEARCH TASK 6

Constructing and Evaluating Synthetic Media

From Data Analysis to Synthetic Audio and Avatar Video

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SYNTHETIC MEDIA DISCLOSURE

All audio and video artifacts described in this report were generated with AI for academic research. No real person's identity, face, or cloned voice was used.

1. Project Overview

This project investigates how a verified, data-driven narrative changes when it is transformed into synthetic media. I created and evaluated two synthetic-media pipelines: an audio-only narration using ElevenLabs and an AI-avatar video using Vidnoz AI. Each pipeline was iterated at least twice, evaluated for realism and failure modes, and documented. I then tested the final ElevenLabs artifact with an automated AI voice detector.

The source narrative was grounded in descriptive analysis of three 2024 presidential-election social-media datasets. The purpose was not to persuade an audience politically, but to examine synthetic-media quality, transparency, detection, and provenance using factual statistical material.

Research Questions

- How realistic are current free-tier synthetic audio and avatar-video tools?
- Which characteristics make synthetic media feel human or artificial?
- Do simple prompt/script changes improve realism?
- Can an automated detector correctly identify a synthetic artifact with known provenance?
- What safeguards remain necessary when synthetic media becomes difficult to detect?

2. Source Data and Verified Ground Truth

Dataset	Rows	Columns
Facebook Ads	246,745	41
Facebook Posts	19,009	56
Twitter/X Posts	27,304	47
Combined	293,058	-

Verified statistics used in the synthetic-media scripts included: Facebook Ads estimated spend of \$261,868,355 and estimated impressions of 11,251,948,521; Facebook-post median interactions of 133 and mean interactions of approximately 2,210 for records with numeric interaction data; and Twitter/X median likes of 1,406 and mean likes of approximately 6,914 across 27,304 posts.

Interpretive limitation: the advertising and post datasets contain different measures. I did not treat ad spending as causing post engagement, and I did not claim a record-level linkage that was not established by the data.

3. Source Script

The full ElevenLabs narration was designed to remain under two minutes and included explicit synthetic-media disclosure. Its core message was that averages can misrepresent typical social-media engagement when a small number of highly engaged posts raise the mean. The script also emphasized that realistic AI presentation does not guarantee factual accuracy and that underlying data must still be verified by humans.

Opening disclosure: “This is an AI-generated synthetic voice, created for an academic research project.”

Closing disclosure: “This presentation was created using artificial intelligence for educational research. No real person’s identity or voice was represented.”

4. Approach 1 - ElevenLabs Synthetic Audio

4.1 Tool and Artifact

Item	Details
Tool	ElevenLabs
Account	Free tier
Voice	Siren - Natural realistic conversational voice
Version 1 duration	1:57
Version 2 duration	approximately 1:56
Final selection	Version 1

4.2 Version 1 - Baseline

Characteristic	Human Rating
Voice naturalness	95%
Pronunciation of numerical statistics	100%
Pacing	100%
Pauses between ideas	80%
Emotional variation	90%

Version 1 was highly convincing in pronunciation, pacing, and general voice quality. The primary synthetic cue was the absence of natural breathing and the limited length of pauses between some sentences. The narration could continue for long stretches without the breathing rhythm expected from a human speaker.

4.3 Version 2 - Pause-Focused Iteration

Characteristic	Human Rating
Voice naturalness	98%
Pronunciation of numerical statistics	100%
Pacing	90%
Pauses between ideas	90%
Emotional variation	95%

For Version 2, the script formatting was changed to create stronger separation between ideas while retaining the same voice and overall content. The result improved perceived naturalness and emotional variation in some places, but it also sounded faster and made the AI nature of the voice more noticeable. The iteration therefore did not produce an unambiguous improvement.

Key finding: adding more pauses mechanically does not necessarily reproduce natural human breathing. Pause placement, duration, rhythm, and variation matter more than simply increasing the number of breaks.

4.4 Final Audio Selection

Version 1 was selected as the final audio artifact because its overall pacing felt more natural, even though Version 2 received slightly higher ratings for voice naturalness and emotional variation. This choice demonstrates that the “best” artifact depends on the combined listening experience rather than a single score.

5. Approach 2 - Vidnoz Synthetic Avatar Video

5.1 Tool and Artifact

Item	Details
Tool	Vidnoz AI
Plan	Free tier
Avatar	Annie
Voice	Annie (Lifelike)
Resolution	720p
Version 1 duration	46.1 seconds
Version 2 duration	51.4 seconds
Final selection	Version 1

5.2 Version 1 - Baseline

Characteristic	Human Rating
Visual realism	80%
Lip synchronization	90%
Facial expressions	50%
Body/head movement	90%
Voice naturalness	80%
Pacing	70%
Emotional variation	60%
Overall realism	80%

Version 1 performed best in lip synchronization and body/head movement. However, the avatar was still prominently recognizable as AI-generated. Facial expression was the weakest visual dimension, and the voice lacked natural emotional variation. The central problem was robotic vocal emotion combined with limited facial expression.

5.3 Version 2 - Short-Sentence Iteration

Version 2 used shorter sentences and stronger separation between ideas while preserving the same research content. I perceived Version 2 as less realistic than Version 1. The longer and more

noticeable pauses disrupted pacing rather than making the avatar appear to breathe or think naturally.

Key finding: more explicit pauses can reduce realism when they become predictable or overly long. Human speech contains irregular rhythm, variable sentence length, breathing, emphasis, and hesitation that a simple sentence-breaking strategy does not reproduce.

5.4 Final Video Selection

Version 1 was selected as the stronger Vidnoz artifact because its continuous pacing was more natural. The experiment also showed that strong lip synchronization alone is insufficient for full realism when facial expression and emotional delivery remain weak.

6. Cross-Approach Evaluation

Dimension	ElevenLabs Audio	Vidnoz Video
Strongest feature	Natural voice and accurate pronunciation	Lip synchronization and body/head movement
Main weakness	Breathing and pause rhythm	Facial expression and robotic vocal emotion
Effect of iteration	Pause changes did not clearly improve realism	Shorter sentences/longer pauses reduced realism
Final artifact	Version 1	Version 1

A shared result emerged across both pipelines: synthetic-media realism depends heavily on timing and rhythm. In both experiments, deliberate attempts to manipulate pauses made some artificial characteristics more noticeable instead of eliminating them. This suggests that human-like delivery requires more than inserting silence; it requires variable breathing, emphasis, facial behavior, and conversational rhythm.

7. Detection and Provenance Check

7.1 Detector Result

Field	Result
Detection tool	AI Voice Detector
Artifact	ElevenLabs Version 1 MP3
Known provenance	100% AI-generated in ElevenLabs
AI score	16%
Verdict	Likely Human
Segments analyzed	19
Maximum segment AI score	50%
Date analyzed	October 1, 2026

The detector produced a false negative. Although the artifact was known to be completely synthetic, the system classified it as “Likely Human” with an average AI score of only 16%. Only the 00:24-00:30 segment reached 50% and was labeled “Likely AI.”

7.2 Interpretation

This result was one of the most important findings of the project. It shows that automated detection can be confidently wrong even when the true origin of an artifact is known. A detector’s “human” verdict should therefore not be treated as proof of authenticity.

The experiment also highlights the distinction between detection and provenance. Detection attempts to infer origin from the finished media. Provenance documents how the media was actually created. In this project, the source script, generation tools, iterations, output files, human evaluations, and detector result provide stronger evidence of origin than the detector score alone.

7.3 Detector Segment Summary

Time	AI Score	Verdict
00:00-00:06	2%	Human
00:06-00:12	25%	Likely Human
00:12-00:18	18%	Likely Human
00:18-00:24	22%	Likely Human
00:24-00:30	50%	Likely AI
00:30-00:36	17%	Likely Human
00:36-00:42	21%	Likely Human
00:42-00:48	4%	Human
00:48-00:54	19%	Likely Human
00:54-01:00	29%	Likely Human
01:00-01:06	4%	Human
01:06-01:12	16%	Likely Human
01:12-01:18	12%	Human
01:18-01:24	14%	Human
01:24-01:30	23%	Likely Human
01:30-01:36	9%	Human
01:36-01:42	3%	Human
01:42-01:48	3%	Human
01:48-01:54	15%	Human

8. Critical Evaluation

What Held Up Well

- ElevenLabs produced highly natural speech and pronounced large numerical statistics accurately.
- Vidnoz achieved strong lip synchronization and convincing body/head movement for a free-tier avatar workflow.

- Both tools could communicate verified statistical findings in accessible formats.
- Explicit spoken/written disclosure could be incorporated into the artifacts.

What Failed or Felt Synthetic

- ElevenLabs did not consistently reproduce natural breathing and human pause rhythm.
- Vidnoz facial expressions were substantially less convincing than its lip synchronization.
- Vidnoz vocal emotion was robotic relative to the visual movement.
- Simple pause manipulation did not reliably improve either pipeline.
- The automated detector failed to identify the known synthetic audio correctly.

Would the Artifact Fool a Viewer or Listener?

The ElevenLabs audio was the more convincing artifact. My human evaluation rated voice naturalness at 95%, and the automated detector classified it as likely human. A listener who encountered the file without its disclosure or provenance could plausibly mistake it for human narration. The Vidnoz video was easier to recognize as synthetic because the avatar's facial expression and emotional delivery were less natural.

9. Ethical and Responsible-Use Measures

- No real person's face or cloned voice was used.
- The scripts explicitly identified the media as AI-generated.
- The statistical claims were grounded in computed dataset values.
- Unsupported causal claims between advertising spending and social engagement were avoided.
- Both successful and unsuccessful iterations were documented rather than presenting only the strongest output.
- The detector result was reported as a fallible signal, not proof of authenticity.

10. Reflection

Before this experiment, I expected improved pause control to make synthetic media feel more human. Instead, both pipelines showed that timing changes can make artificial characteristics more noticeable. Human speech and presentation involve irregular breathing, emphasis, emotional expression, facial movement, and pacing that are difficult to reproduce through simple formatting changes.

The detection experiment was even more surprising. The final ElevenLabs artifact was completely synthetic, yet the detector assigned it only a 16% AI score and labeled it likely human. This changed my view of detection systems: they can be useful signals, but they should not be treated as final authorities on authenticity.

The strongest verification approach is therefore layered: clear disclosure, documented provenance, verified source material, human review, and automated detection used as supplementary evidence.

11. Conclusion

This project demonstrates that free-tier AI tools can already create synthetic media with substantial realism. ElevenLabs produced an audio artifact that was realistic enough to receive a “Likely Human” detector verdict, while Vidnoz produced strong lip synchronization but remained visually identifiable as synthetic because of facial-expression and emotional limitations.

The project also demonstrates that iteration does not guarantee improvement. In both pipelines, changes intended to improve pauses or pacing introduced new synthetic cues. The central lesson is that realistic synthetic media requires more than technical generation quality: responsible use also depends on factual verification, disclosure, provenance, and human judgment.

12. Repository Contents for Submission

The public GitHub repository should be named: Task_06_Deep_Fake

Path	Purpose
README.md	Project overview, tools, findings, and navigation
SCRIPT.md	Source narration/script with synthetic-media disclosure
PROCESS_LOG.md	Tools, settings, iterations, failures, screenshots, and time log
EVALUATION.md	Critical comparison of ElevenLabs and Vidnoz artifacts
DETECTION_REPORT.md	AI Voice Detector result and provenance discussion
artifacts/elevenlabs_iteration_1.mp3	Baseline/final ElevenLabs audio
artifacts/elevenlabs_iteration_2.mp3	Second ElevenLabs iteration
artifacts/vidnoz_iteration_1.mp4	Baseline/final Vidnoz video
artifacts/vidnoz_iteration_2.mp4	Second Vidnoz iteration
evidence/	Screenshots of generation interfaces and detector result

13. Final Submission Checklist

- Create a public GitHub repository named Task_06_Deep_Fake.
- Upload the documentation files listed above.
- Upload both ElevenLabs MP3 iterations and both Vidnoz MP4 iterations.
- Clearly label the final artifacts as synthetic in the README and within the media where possible.
- Add screenshots showing tool/interface settings, generation steps, and the AI Voice Detector result.
- Check that the repository opens without requiring permission and that media files can be accessed.
- Submit the public GitHub repository URL through the course submission workflow.

- Complete the mandatory Qualtrics questionnaire required by the Task 6 assignment.
- Keep the repository public until grading is complete.

Appendix A - Short Vidnoz Script

AI-GENERATED SYNTHETIC MEDIA - ACADEMIC DEMONSTRATION

This is an AI-generated synthetic presenter created for academic research. For this project, I analyzed social-media data associated with the 2024 presidential election. One important pattern appeared in engagement. For Facebook posts, the median was 133 interactions, while the mean was approximately 2,210. Twitter/X showed a similar pattern, with a median of 1,406 likes compared with a mean of approximately 6,914. This demonstrates how highly engaged posts can increase averages and potentially distort our understanding of typical engagement. AI can communicate these findings convincingly, but realistic presentation does not guarantee accuracy. This synthetic presentation was created for educational research.

Appendix B - Evidence to Add to GitHub

- Screenshot of ElevenLabs Version 1 voice/model/settings.
- Screenshot of ElevenLabs Version 2 settings or changed script formatting.
- Screenshot of Vidnoz avatar Annie and Annie (Lifelike) voice selection.
- Screenshot of Vidnoz generation interface.
- Screenshot of AI Voice Detector showing 16% AI / Likely Human.
- Optional screenshot or still frame from each final artifact.

Appendix C - Source Note

The descriptive statistics in this report were calculated from the three CSV files supplied for this project. The generation/evaluation observations were recorded during the user's actual ElevenLabs and Vidnoz experiments on October 1, 2026. The detector results were transcribed from the AI Voice Detector output supplied after testing the final ElevenLabs Version 1 artifact.